

Problem Sheet 1

1. Suppose that Z_1, Z_2, Z_3 are zero mean random variables with

$$\begin{aligned}\text{var}(Z_1) &= 1, \text{var}(Z_2) = 2, \text{var}(Z_3) = 3, \\ \text{cov}(Z_1, Z_2) &= -0.5, \text{cov}(Z_2, Z_3) = 2.5, \text{cov}(Z_1, Z_3) = 0.\end{aligned}$$

Calculate each of the following:

- $E(Z_1^2 - Z_2 - Z_2 Z_3)$
 - $\text{var}(2Z_1 + 3Z_2 - Z_3)$
 - $\text{cov}(3Z_1 - Z_2, Z_2 - 2Z_3)$
 - $\text{corr}(Z_1, 2Z_2 + Z_3)$
2. Suppose $\{e_t\}$ is a normal white noise process with mean zero and variance σ_e^2 . Let $\{Y_t\}$ be a process defined as:
- $$Y_t = e_t + \theta e_{t-1}.$$
- Find the autocovariance function and autocorrelation function of Y_t for any general θ . [Hint: Calculate $\text{cov}(Y_t, Y_{t-k})$ case-by-case for several values of k .] Also, find the autocovariance function and autocorrelation function of Y_t if $\theta = 2$.
 - Is the time series $\{Y_t\}$ stationary?.
3. Apply a moving average filter to Y_t , where Y_t is the natural logarithm of the Johnson and Johnson earnings data (the original data are given in the `jj` object in the `astsa` package). Specifically, let

$$V_t = \frac{Y_t + Y_{t-1} + Y_{t-2} + Y_{t-3}}{4}.$$

The R code

```
v = filter(y, rep(1/4, 4), sides = 1)
```

may be helpful in implementing this. Type `help(filter)` in R for more details about this R function. Plot Y_t as a line and overlay (superimpose) V_t as a dashed line, and provide this plot. Discuss whether the moving average filter captures the overall trend in the time series.

4. Construct a time series plot of the Dubuque temperature data, but include the monthly plotting symbols. Note that the temperature data and the month information are in the `tempdub` object in the `TSA` package. Type `library(TSA); data(tempdub); print(tempdub)` in R to see the data set.
5. (a) Simulate and plot a white noise process $e_t \stackrel{\sim}{\sim} \text{iid } N(0, 1)$ of length $n = 100$ using the following commands in R:
- ```
> wn.n01 = rnorm(100,0,1)
> plot(wn.n01,ylab="White noise process",xlab="Time",type="o")
```
- (b) Repeat part (a) under the assumption that
- $e_t \stackrel{\sim}{\sim} \text{iid } t(1)$
  - $e_t \stackrel{\sim}{\sim} \text{iid } \chi^2(4)$

To do this, just replace the first line of the code above with `wn.t1 = rt(100,1)` and `wn.chisq4 = rchisq(100,4)`, respectively. Comment on the differences among the 3 simulated white noise processes.

- (c) Repeat part (a) using  $n = 200$ ,  $n = 300$ , and  $n = 500$ . With your plot from part (a), take your 4 standard normal white noise processes and put them in a  $2 \times 2$  matrix of plots using the `—par(mfrow=c(2,2))—` command in R. Label each plot in the matrix according to the sample size used, e.g.,

```
plot(wn.n01.100,ylab="WN",xlab="Time",main="Sample.size=100",type="o"):
```

6. Suppose that  $Z_1$  and  $Z_2$  are uncorrelated random variables with zero mean and unit variance. Consider the process defined by

$$Y_t = Z_1 \cos(\omega t) + Z_2 \sin(\omega t) + e_t,$$

where  $e_t \stackrel{\text{iid}}{\sim} N(0, \sigma_e^2)$  and  $\{e_t\}$  is independent of both  $Z_1$  and  $Z_2$ .

- (a) Prove that  $\{Y_t\}$  is stationary.  
 (b) Let  $Z_1$  and  $Z_2$  be independent  $N(0,1)$  random variables, and set  $\sigma_e^2 = 1$  and  $\omega = 0.5$ . Use the following R commands to simulate  $n = 150$  observations from the  $\{Y_t\}$  process:

```
> omega = 0.5
> Z = rnorm(2,0,1)
> e.t = rnorm(150,0,1)
> Y.t = e.t*0
> for (i in 1:length(e.t)){Y.t[i] = Z[1]*cos(omega*i)
+ Z[2]*sin(omega*i) + e.t[i]}
> plot(Y.t,ylab="Trigonometric process",xlab="Time",type="o")
```

Describe the appearance of your time series.

- (c) Amend the R code above to simulate a realisation of the process

$$\tilde{Y}_t = \beta_0 + \beta_1 t + Z_1 \cos(\omega t) + Z_2 \sin(\omega t) + e_t,$$

where  $\beta_0 = 100, \beta_1 = 0.05, \sigma_e^2 = 1$  and  $\omega = 0.5$ .

Does your  $\{\tilde{Y}_t\}$  process appear to be stationary? What is the effect of adding the linear trend term  $\beta_0 + \beta_1 t$  to the model?

- (d) Plot the first differences of your simulated  $\{\tilde{Y}_t\}$  process. To do this, use

```
> diff.Y.tilde = diff(Y.tilde)
> plot(diff.Y.tilde,ylab="First differences of Y.tilde",xlab="Time",type="o")
```

Describe the appearance of this first difference process  $\{\nabla \tilde{Y}_t\}$ . In particular, does it appear to be stationary in the mean level? Are you surprised? Discuss the behavior of each process:  $\{Y_t\}$ ,  $\{\tilde{Y}_t\}$ , and  $\{\nabla \tilde{Y}_t\}$ , and how they relate to each other.

- (e) In part (c), suppose that instead of the added linear trend  $\beta_0 + \beta_1 t$ , the added trend was quadratic, say,  $\beta_0 + \beta_1 t + \beta_2 t^2$ . Do you think the first differences of the quadratic trend version of the process in part (c) would be stationary? Explain your intuition.

7. Suppose  $\{e_t\}$  is a normal white noise process with mean zero and variance  $\sigma_e^2$ . Let  $\{Y_t\}$  be a process defined as:

$$Y_t = e_t e_{t-1}.$$

Showing all your steps, find the mean function and the autocovariance function of  $Y_t$ . Is the time series  $\{Y_t\}$  stationary? Explain your answer.